

Class: BE-Computer; Semester: VIII
Subject: Applied Data Science Lab
Experiment No.: 07

AIM-Anomaly/ Outlier detection techniques

Theory

- Types of outliers -Global outliers ,Contextual outliers, Collective outliers, give example of each.
- Why outlier detection is important? -
 - Outliers can skew trends and forecasts modelled from datasets, negatively impacting the quality and accuracy of decisions.
 - Prevention Fraudulent attacks

Method

- 1) Visual analysis of the Univariate data to observe extreme values in the data.
 - Distplot,Boxplots, Scatter plots, Whisker plots, etc.,
- 2) Distance based outlier detection
 - K-NN based outlier detection
- 3) Density based outlier detection (implement any one)
 - Local outlier Factor (LOF)
 - DB SCAN
- 4)Remove outliers using trimming and capping(Winsorization)
- 5)Compare the distribution and box-plot after capping

Implementation Details

Dataset used

Library

Pycaret.anomaly
sklearn.cluster.DBSCAN
sklearn.neighbours.NearestNeighbours